EDA Report:

Contents

Executive summary: 2

Data Overview and Pre-processing: 3

Overview of the data: 3

Data pre-processing: 4

Crime by Police Force Area: 8

LSOA level analysis: 13

Conclusion: 15

Executive summary:

Essex, Hertfordshire, Kent and Surrey occupy a desirable position in the U.K. property market as four principal commuter belt counties surrounding London. Offering a combination of relative affordability, transport links and access to green spaces these counties consistently attract strong demand from buyers seeking to balance quality of life with proximity to work.

This report presents an Exploratory Data Analysis of recorded crime across four Police Force Areas (PFAs) in the London commuter belt – Essex, Hertfordshire, Kent and Surrey – over the period 2022 to 2024. The analysis mainly sources data from data.police.uk, with various geographical and population data sourced from the ONS. The findings are intended to brief and support a real estate sales team on the most and least desirable areas to make more informed decisions around sales.

At the PFA level, Kent and Essex consistently record the highest overall crime rates while Hertfordshire and Surrey sit notably lower. This pattern persists across the violent and sexual crime category. Surrey emerges clearly as the most uniformly low-crime PFA in the report, with Hertfordshire offering similarly low figures outside of unexpectedly high anti-social behaviour rate. Burglary rates are consistent between PFAs. Both Kent and Essex show a significant downward trend in violence and sexual offences and burglary rates between 2022 and 2024.

At the LSOA level, a pronounced urban-rural divide is evident across all four PFAs, with crime hotspots concentrated in town centres such as Canterbury (Kent), Southend-on-Sea (Essex), Gravesham (Kent), and Dartford (Kent), while rural and suburban LSOAs extremely low crime rates.

Overall, Surrey and Hertfordshire represent the strongest proposition in terms of crime. Targeting the lowest crime rate LSOAs in these counties represent the lowest crime risk for clients looking to purchase

Data Overview and Pre-processing:

Overview of the data:

Police forces data:

(data.police.uk/data/archive)

Crime data was obtained from data.police.uk/data/archive, covering a three-year period spanning January 2022 to December 2024. The data used was from the Essex, Hertfordshire, Kent and Surrey police forces. The data contains columns denoting each crime reported by the respective police force, the anonymised coordinates of the crime, the LSOA code and name, the crime type and the last outcome category.

LSOA population data:

(ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/lowersuperoutputareamidyearpopulationestimatesnationalstatistics)

The population data for each LSOA was sourced from the ONS. The data denotes the mid-year population estimates of the usual resident population for Lower Layer Super Output Areas in England and Wales.

PFA population data:

(ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/adhocs/3194populationestimatesforpoliceforceareasinenglandandwalesbysingleyearofageandsexmid1991tomid2024?utm_source=copilot.com)

The population data for each police force area was sourced from the ONS. The data denotes the mid-year population estimates of each area.

PFA boundaries data:

(data.police.uk/data/boundaries)

The boundaries data for each police force area was sourced from data.police.uk/data/boundaries. The data gives the geometry data for each area.

LSOA boundaries data:

(geoportal.statistics.gov.uk/datasets/ons::lower-layer-super-output-areas-december-2021-boundaries-ew-bsc-v4-2/about)

The boundaries data for each LSOA was sourced from the ONS' open geography portal. The data gives the .shp file containing the geometry data for each LSOA.

Data pre-processing:

The data was pre-processed in multiple stages: loading the data, auditing the data, cleaning the data, combining the data and exporting the data. The flowchart should describe the approach to pre-processing that was taken. This section gives a more detailed description of the pre-processing. By the end of pre-processing, around 1% of records in the original crime dataset was dropped.

Loading the data:

Police forces data:

Firstly, the empty DataFrame for each police force were created. Then, four empty lists were created for each police force, then populated filenames containing the relevant force name into the appropriate. Each police force's individual CSV files were read and concatenated, producing a single consolidated DataFrame per force containing all their respective records.

Other data:

The other data was loaded using pandas and geopandas functions without any difficulty.

Auditing the data:

To audit the dataset, the .shape and .info() functions were used to determine the number of rows and columns and to check for any null values. The .head() function was then used to visually inspect the data and identify any other unusual patterns or inconsistencies.

Police forces data:

The main issues found were completely null column 'Context', null location and coordinate data and null 'Crime ID' and 'Last Outcome Category' data.

Further investigation showed that the 'Crime ID' and 'Last Outcome Category' data were null because 'Anti-social behaviour' isn't categorised as a crime.

LSOA population data:

The main issue found was that the dataset contained a lot of irrelevant data which could be dropped to make the dataset more lightweight.

PFA population data:

The main issues found were that the dataset contained a lot of irrelevant data which could be dropped to make the dataset more lightweight, the number of columns was too large, and there was no 'Total' population column.

LSOA boundary data:

The main issues found were that the dataset held many columns that were irrelevant to the analysis and the column 'LSOA21NMW' held null values and wasn't useful, so could be dropped.

PFA boundary data:

The main issues found were that the dataset held many columns that were null or irrelevant to analysis that could be dropped and that the 'Name' column was null for each boundary.

Cleaning the data:

Police forces data:

The context column was dropped. The nulls in 'Crime ID' and 'Last Outcome category' were filled with 'Not crime' and 'No data' respectively. The rows with no location data were removed. The 'Month' column was split into 'Months' and 'Years'. A 'PFA' column was inserted, to denote which PFA each record belonged to.

LSOA population:

A check was done to select all the LSOAs in the population data that were in the police forces data. Doing this, it was found that many records in the police forces data referenced LSOAs from outside their respective PFAs. To rectify this, those LSOAs from outside the PFAs were dropped. Then the unnecessary LSOAs in the population data could be dropped. A column representing the year for each of the datasets was added. Then the LAD codes were dropped as they were not needed during the analysis.

PFA population:

The rows that didn't relate to the PFAs for Essex, Hertfordshire, Kent and Surrey were dropped. The columns for population were combined into buckets that mirrored the LSOA population data. A new column 'Total' was created that summed the population data for each PFA.

LSOA boundaries:

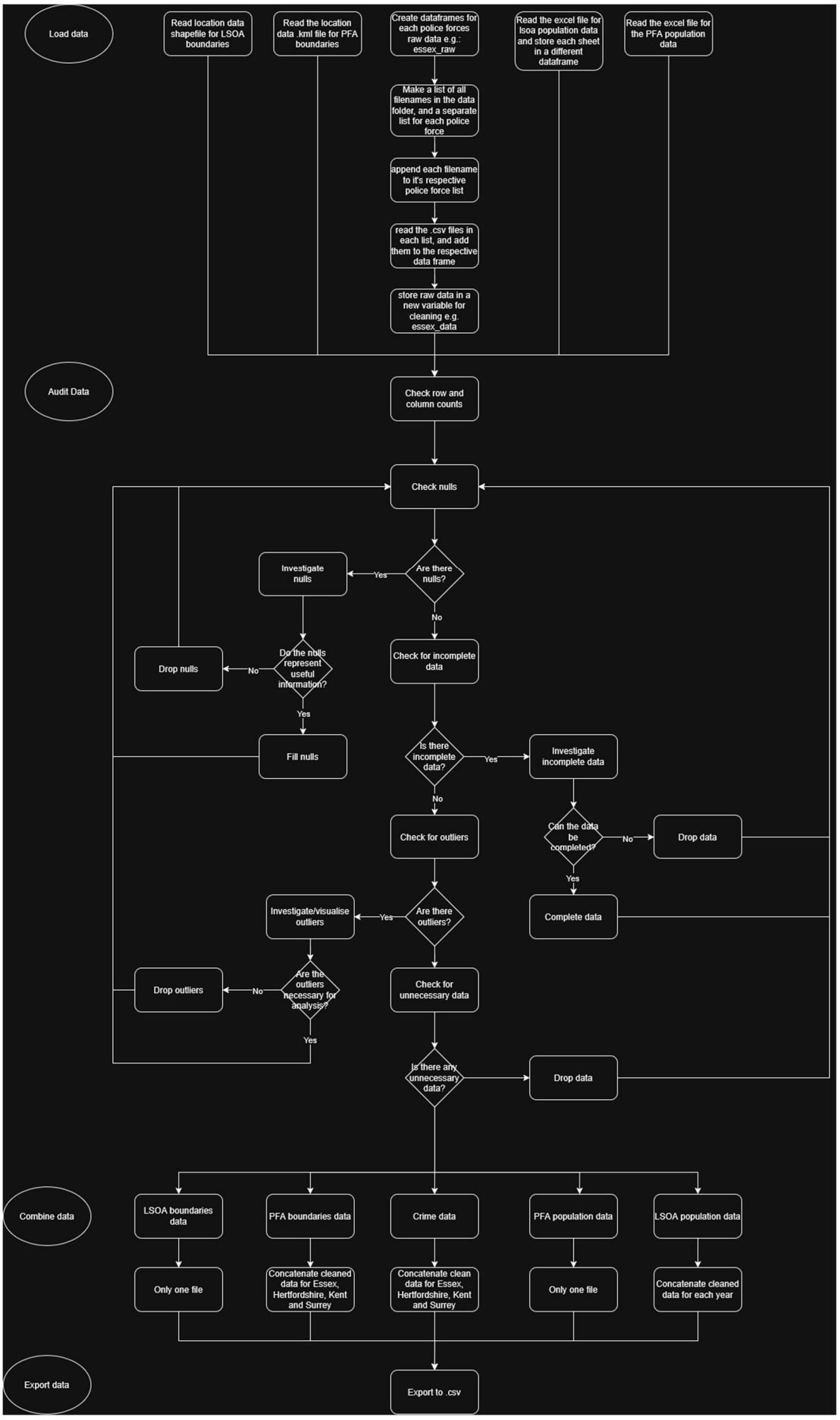
The unnecessary LSOAs were dropped.

PFA boundaries:

Each PFA was given a name value. The unnecessary columns in each PFA boundary dataset were dropped.

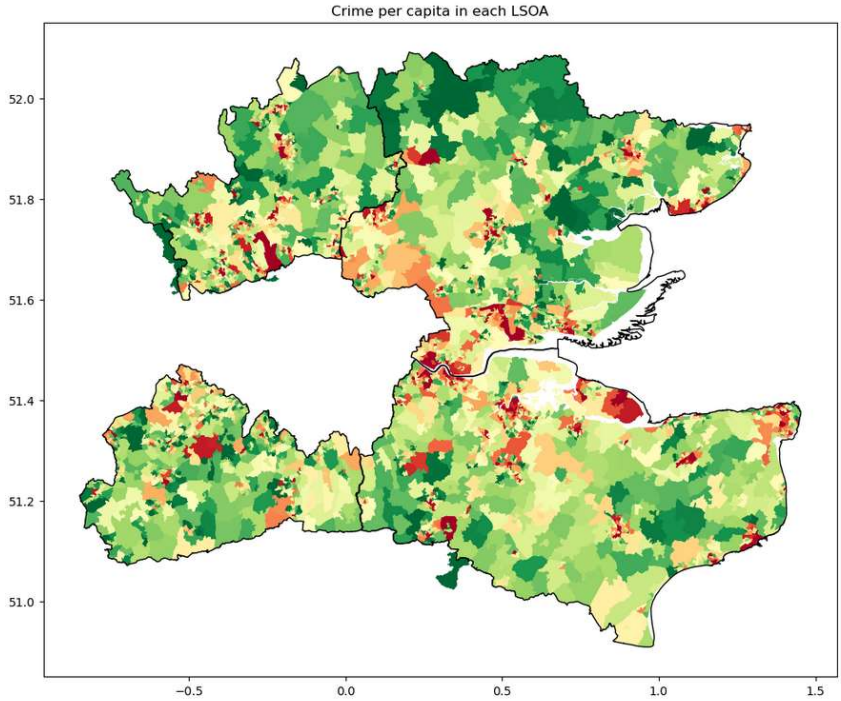
Combining and exporting data:

Each dataset that was divided into multiple dataframes was combined with their respective datasets to create a single dataframe that could be exported. The datasets were exported as .csv to be used for the EDA.

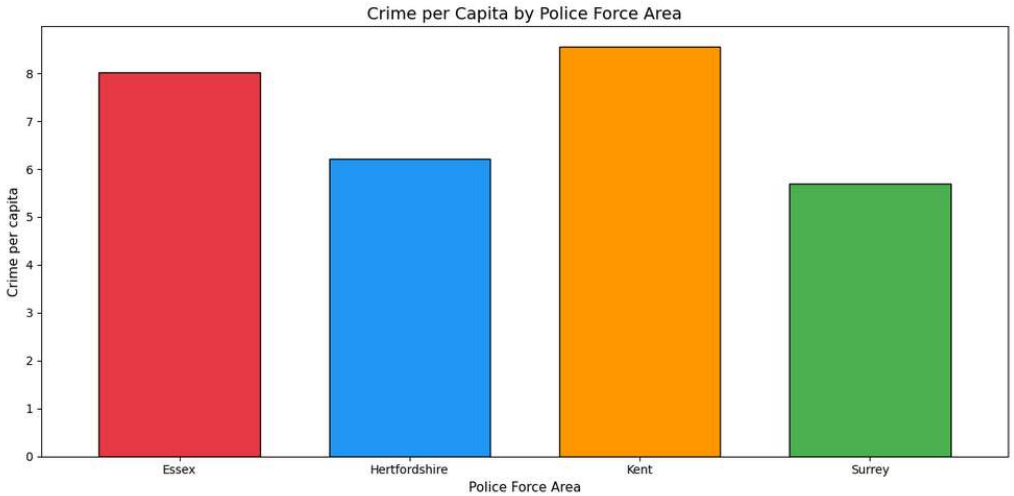


Crime by Police Force Area:

Overall Crime:



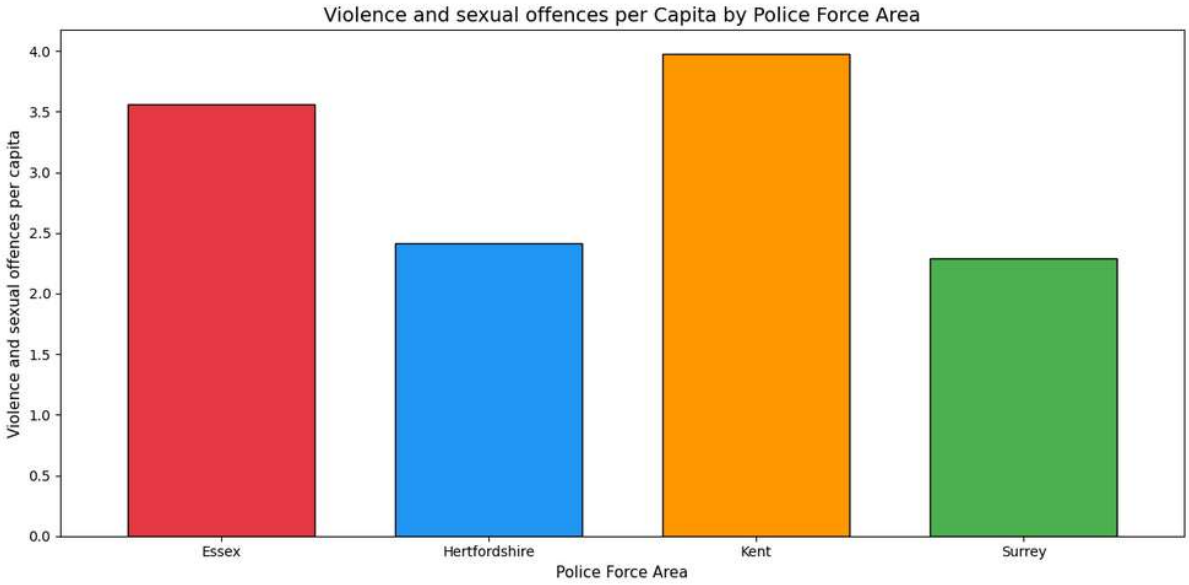
The choropleth map above displays the geographic distribution of crime per capita across all LSOAs within the four PFAs: Essex, Hertfordshire, Kent, and Surrey. The colour scale ranges from dark green (low crime per capita) to dark red (high crime per capita). A notable pattern shown is that most crime is concentrated within urban centres, most visibly in the south of Hertfordshire, the south of Essex and the north of Kent. Contrastingly more rural LSOAs, particularly Surrey, Northern Hertfordshire and Northern Essex display the lowest crime rates.



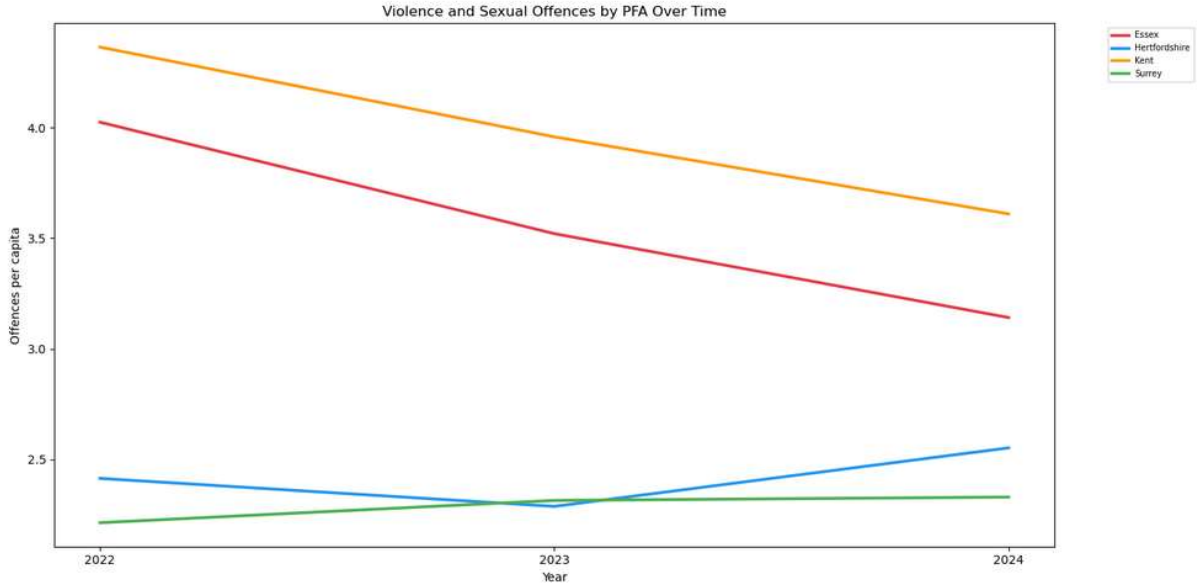
The bar chart above shows a comparison of the overall crime per capita across the four PFAs. Kent and Essex record the highest crime rates while Hertfordshire and Surrey exhibit the lowest. This chart reveals a meaningful variation in crime levels across the

region despite their geographic proximity. The immediate takeaway from this graph is that Kent and Essex exhibit significantly higher crime rates than Hertfordshire and Surrey.

Violent and Sexual Offences:

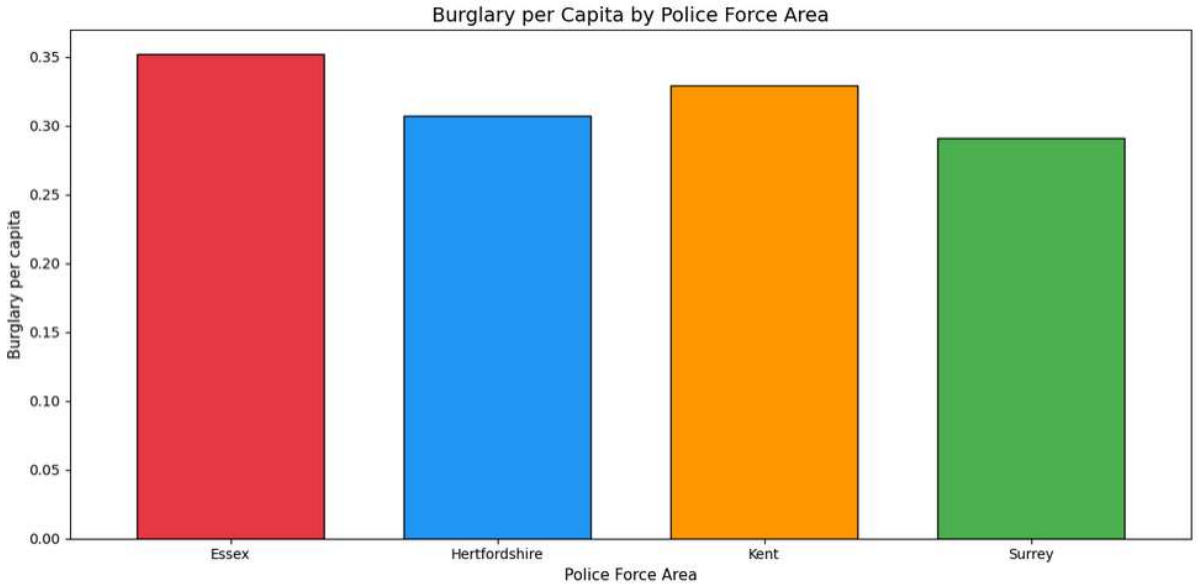


Kent records the highest rate of violence and sexual offences per capita at approximately 3.9, making it notably higher than the other three PFAs. Essex follows at around 3.6, also considerably above the group average. Hertfordshire and Surrey record similar and substantially lower rates, continuing the pattern observed in the overall crime per capita chart.

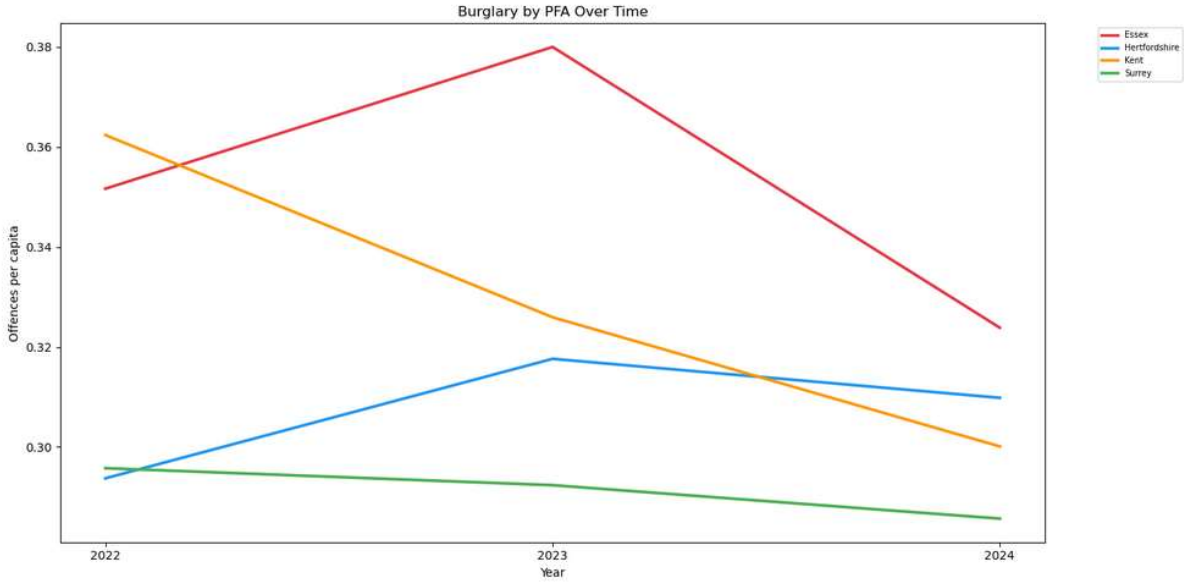


Unsurprisingly, Kent and Essex both begin with notably higher rates (approximately 4.4 and 4.0 respectively in 2022) and decline steadily to around 3.6 and 3.1 by 2024, suggesting a meaningful reduction in recorded violent crime across both forces. Hertfordshire and Surrey, by contrast, remain consistently lower and relatively stable across the period.

Burglary:



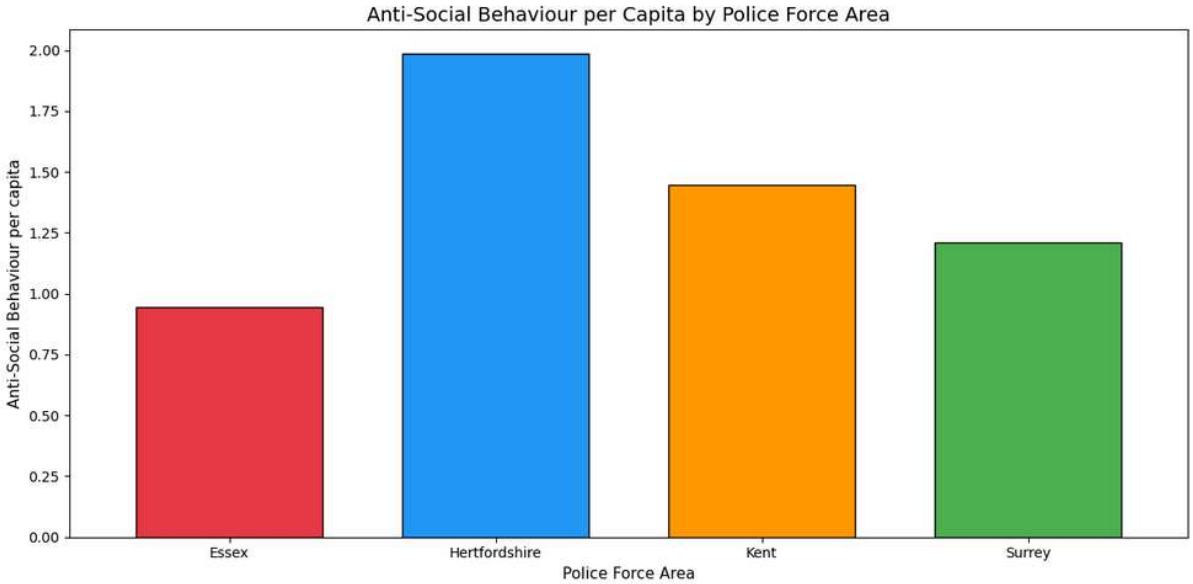
In contrast to the overall crime and violence figures, the distribution of burglary rates across the four PFAs is notably more uniform. Essex records the highest burglary rate at approximately 0.35 per capita, followed closely by Kent at around 0.33. Hertfordshire sits at approximately 0.30, with Surrey recording the lowest rate at around 0.29. The difference between the highest (Essex) and lowest (Surrey) is only approximately 0.06 per capita - a considerably narrower gap than that observed for total crime or violence and sexual offences.



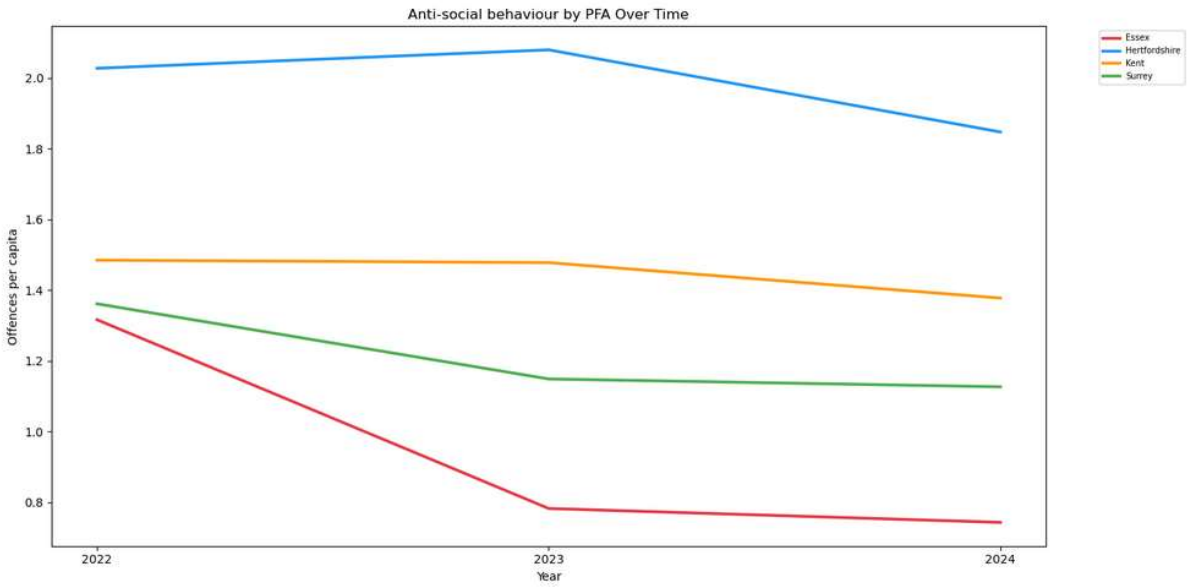
Over the period of 2022 to 2024 burglary rates in the respective regions have changed drastically. In 2022, Kent begins as the highest but declines sharply, displaying the most pronounced reduction in burglary across the PFAs. Essex increases sharply in 2023 but,

similarly, drops drastically by 2024. Contrastingly, Hertfordshire increases overtime. Burglary rates in Surrey stay modestly low during the period.

Anti-Social Behaviour:



Unexpectedly, when it comes to anti-social behaviour, Hertfordshire displays the highest rates, and Essex shows the lowest rates. Although not technically a category of crime, anti-social behaviour could still affect the desirability of property in the respective regions. This result reflects another story that was not represented by the overall crime statistics.

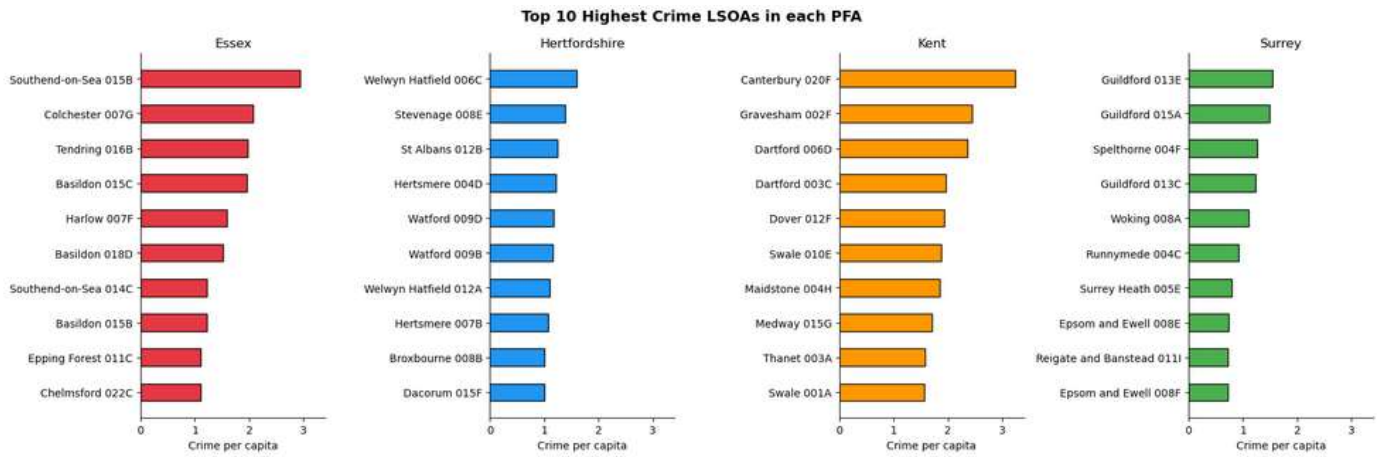


Hertfordshire remains the clear outlier throughout, starting at approximately 2.02 in 2022, peaking slightly at around 2.05 in 2023, before declining to approximately 1.85 by 2024, remaining well above the other three PFAs across the entire period. Kent declines

modestly from approximately 1.49 to 1.38, while Surrey falls from around 1.36 to 1.14. Essex shows the steepest decline of any PFA, dropping sharply from approximately 1.32 in 2022 to around 0.76 by 2024.

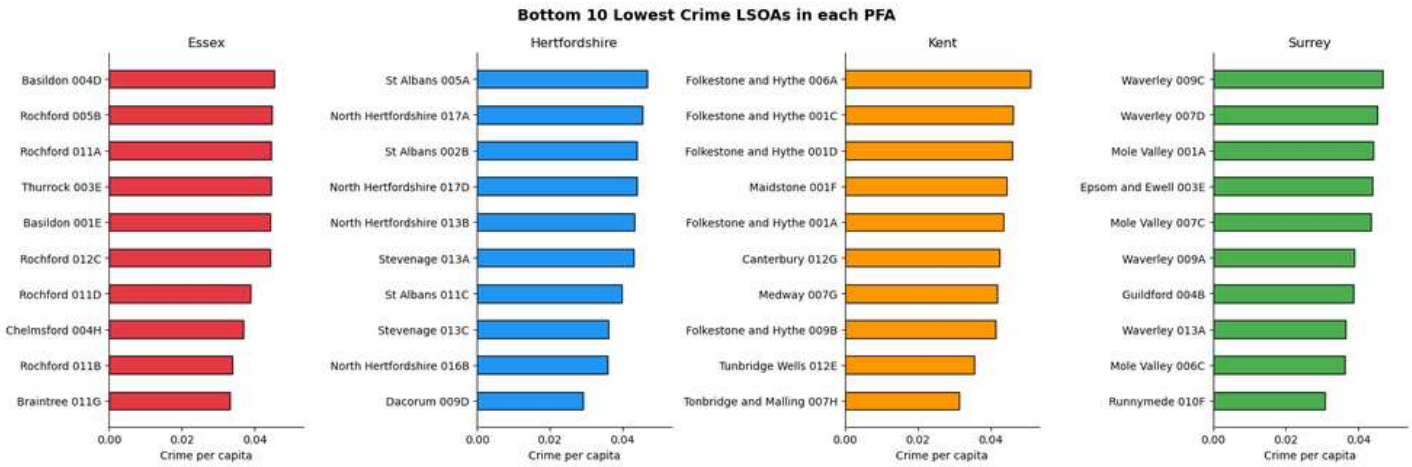
LSOA level analysis:

Most dangerous LSOAs per PFA:



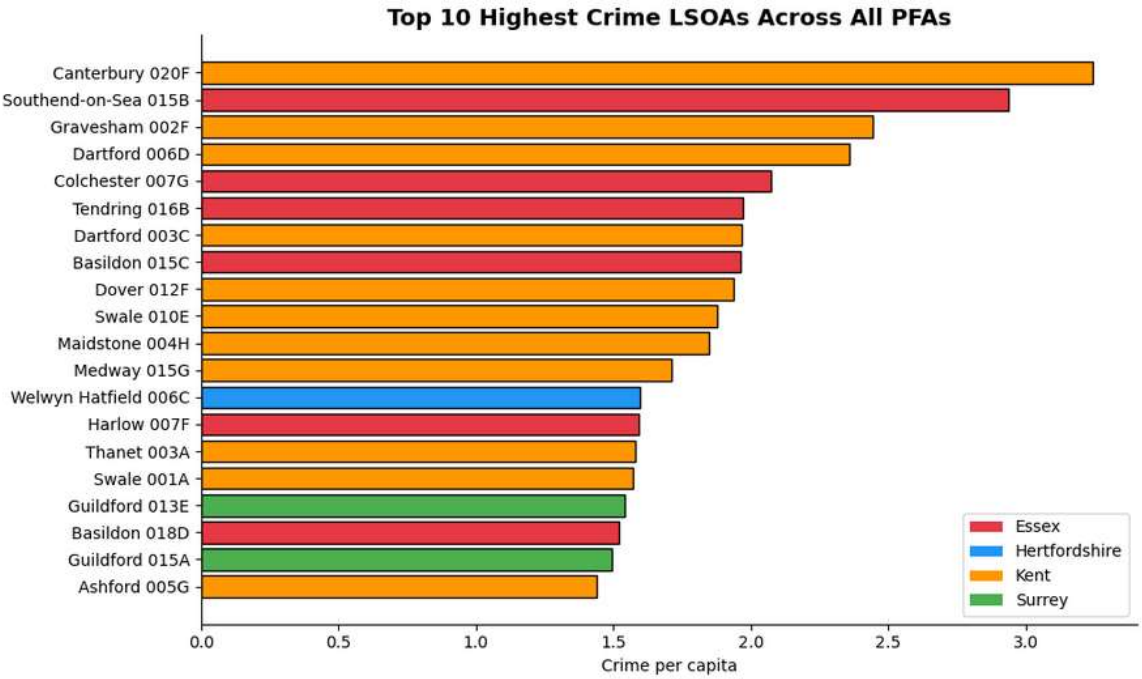
This chart shows the 10 LSOAs with the highest crime within each PFA. From a real estate perspective, these will be the least desirable areas, in each PFA, to invest.

Least dangerous LSOAs per PFA:



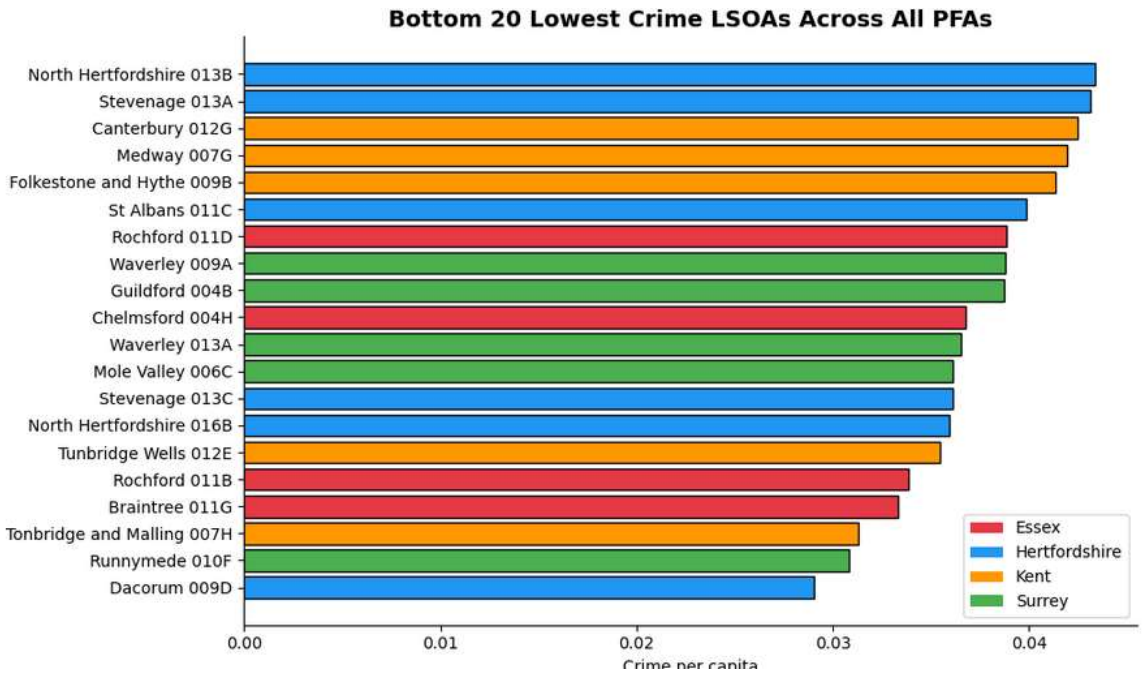
This chart shows the 10 LSOAs with the lowest crime within each PFA. From a real estate perspective, these will be the most desirable areas, in each PFA, to invest.

Most dangerous LSOAs overall:



This chart shows the top 20 LSOAs with the highest crime rate overall. Most of these fall under the Kent and Essex PFA. From a real estate perspective, these will be the least desirable areas, in each PFA, to invest.

Least dangerous LSOAs overall:



This chart shows the top 20 LSOAs with the lowest crime rate overall. These LSOAs come from a mix of PFAs. From a real estate perspective, these will be the most desirable areas, in each PFA, to invest.

Conclusion:

Overall, lower crime rates were displayed in Hertfordshire and Surrey, suggesting an increased desirability for real estate in those areas. More analysis that focuses on these counties would be beneficial for a clearer and more informed decision-making process.

Limitations:

- Police.uk data captures only reported and recorded crime, meaning the true volume of criminal activity in any given area is likely higher than the figures suggest.
- Variation in how each force classifies and records incidents can distort cross-PFA comparisons. This is plausibly seen in the anti-social behaviour figures for Hertfordshire. Some differences between forces may therefore reflect administrative practices rather than genuine differences in underlying crime levels.
- Per capita figures rely on ONS mid-year population estimates, which are modelled projections rather than exact counts.
- The report focuses solely on crime data and does not incorporate property market variables. Crime rates are one component of property desirability and should complement broader market analysis.
- The three-year study period is sufficient to identify broad trends but limits the ability to distinguish sustained structural change from short-term fluctuation. A longer time series would provide greater confidence in the directional trends observed, particularly the declining violence figures in Kent and Essex.